Pre-Compliance EMC at the Bench

Find compliance failures BEFORE the expensive lab visit

Save ₹1-3 L per cert by catching issues in-house

Audience: Hardware engineers + small startups

Near-field probes • LISN • Spectrum analyzer • ESD gun • Methodology

Why Pre-Compliance Matters

Cost savings

Pre-comp scan ₹50-80k
vs full cert ₹3-15 L
(and re-tests)

Time savings

Find issues in days,
not weeks of waiting.

Iterations

Fix → re-scan → fix → scan
vs single shot at lab.

Confidence

Walk into lab knowing
you'll pass.

Continuous

Test every design rev,
not just final.

DIY accessible

Equipment 50k-50L,
rentable elsewhere.

What Pre-Compliance Covers

- Conducted emissions: noise via power cord (150 kHz - 30 MHz)
- Radiated emissions: noise via airwaves (30 MHz - 1 GHz typically)
- ESD immunity: contact + air discharge survival
- Surge / EFT (burst) immunity: lightning + switching transients
- Radiated susceptibility: signal injection at various frequencies
- Power-line dips + interruptions
- RF immunity: external RF field doesn't disrupt operation
- Pre-compliance ≠ full certification — but catches 80%+ of failures

Near-Field Probes

- Small magnetic loop or E-field probe — handheld
- Locates radiation hot spots on a PCB
- Connects to spectrum analyzer via SMA cable
- Move probe around board → watch peak frequencies
- Identifies: SMPS switching, clock harmonics, USB / Ethernet traces
- Cost: ₹15-50k for a probe set (Beehive Electronics, Würth)
- Limitation: relative measurement, not absolute compliance number
- Best for: finding noise SOURCES → then fix at the design level

LISN (Line Impedance Stabilization Network)

- Couples power-line noise from DUT to spectrum analyzer
- Provides defined impedance for repeatable conducted emissions measurement
- Required for any conducted-emissions pre-compliance
- Single-phase LISN ₹30-80k (Schaffner, EMV-Tek)
- Three-phase LISN ₹1-3 L (industrial products)
- Used per CISPR 16 / EN 55016 standards
- Connected: AC mains → LISN → DUT, then LISN sample output → analyzer

Spectrum Analyzer for Pre-Comp

- Frequency span: 9 kHz - 1 GHz minimum (covers most CISPR / EN standards)
- Resolution bandwidth (RBW): 9 kHz / 120 kHz / 1 MHz per CISPR bands
- Quasi-peak (QP) detector — emulates lab measurement
- Display ITU compliance limits as overlay on screen
- Models: Rigol DSA815 (₹1.5L), Siglent SSA3032X-R (₹2L), Keysight ESA
- Software: CISPR pre-compliance scan software (commercial or open-source)
- Tek RSA series — more advanced, real-time, ₹5-10L

ESD Gun

- Simulates human ESD discharge per IEC 61000-4-2
- Apply 8 kV contact + 15 kV air discharge to every accessible surface
- Watch DUT for: reset, lockup, function loss, latching damage
- Consumer-grade gun: ₹40-80k (NoiseKen, EM Test, Schaffner)
- Industrial gun: ₹3-8 L (calibrated, certified)
- Test points: every USB / connector pin, every accessible signal, every external case point
- Document: which points caused which failure

Surge + EFT Tester

- Surge: lightning-equivalent voltage transient (1-6 kV) per IEC 61000-4-5
- EFT/Burst: switching transient bursts (1-4 kV) per IEC 61000-4-4
- Apply to: power lines, signal cables entering the DUT
- DUT must survive without damage; may glitch (depending on Class A/B)
- Compact tester: NoiseKen ESS-2002, ₹4-10 L (used market cheaper)
- DIY surge tester: NOT recommended — high-energy + dangerous
- Often rented at certified labs for ₹30-50k/day

DIY Anechoic Setup

- True anechoic chamber: ₹10-50 Cr (only for major labs)
- DIY semi-anechoic: copper-clad room + RF absorber panels (₹2-5 L)
- Easier: portable RF tent (₹50k-2L) — Würth, Holland Shielding
- Inside tent: spectrum analyzer + DUT + biconical antenna
- Calibrated for relative pre-comp measurements — NOT compliance reports
- Outdoor open-area test site (OATS) — quiet field + portable equipment
- Pre-comp goal: get within 6-10 dB of the limit, then go to certified lab

Typical Pre-Comp Workflow

- 1. Read applicable standard (EN 55032, CISPR 22, FCC Part 15)
- 2. Setup DUT in 'typical operation' mode
- 3. Conducted scan via LISN: sweep + record exceeding limits
- 4. Radiated scan: near-field probe to identify hot spots, then antenna at distance
- 5. ESD gun: apply 8/15 kV to every accessible point
- 6. Document failures + root cause hypotheses
- 7. Apply fixes (CMC, TVS, decoupling, shielding)
- 8. Re-scan → iterate until safely below limits → schedule full cert

Common Pre-Comp Findings

- Conducted emissions @ 1 MHz: SMPS switching → add input filter (CMC + caps)
- Conducted @ 30 kHz: 50/60 Hz harmonics from poor PSU layout
- Radiated @ MCU clock harmonics: missing decoupling, long clock traces
- Radiated @ USB cable: missing common-mode choke at USB connector
- ESD failure at USB-C: missing TVS array on diff pairs
- ESD failure at GPIO header: bare pin, no TVS → MCU damaged
- Surge failure: no MOV or GDT on AC input

Renting vs Buying

- Spectrum analyzer + LISN: BUY if testing > 5 designs/year (pays back fast)
- ESD gun: BUY if you ship hardware (essential, frequent use)
- Surge / EFT tester: RENT — used rarely, expensive
- Anechoic tent: RENT or use partner lab
- Many EMC consultants in India: BENT, Lakshmi Labs, EMC India — rent bench time
- University labs often have equipment — collaboration arrangement
- Cost benchmark: pre-comp consultancy half-day ₹20-50k

Key Takeaways

Key takeaways from this lecture

Pre-comp = cost saver

₹50-80k vs ₹3-15 L for full cert.

Near-field probe

Find emission hot-spots fast.

LISN for conducted

Standard for power-line emissions.

ESD gun = must-have

Apply 8/15 kV to every connector pin.

Document everything

Fail mode → fix → re-test.

Build OR rent

Match volume of designs to investment.

Questions & Discussion

Ask anything — concepts, projects, sourcing, career.

References & Further Learning

- Henry Ott — EMC Engineering (foundational)
- Würth Electronik — Trilogy of Magnetics (free PDF)
- EMCFastPass YouTube — pre-comp tutorials
- Indian EMC labs: BENT, Lakshmi, SGS, Bureau Veritas
- Schaffner / EMV-Tek / NoiseKen pre-comp equipment
- CISPR / EN 55032 / EN 55035 standards (or summaries)
- Robert Feranec / Phil's Lab YouTube
- IEC standards 61000-4-x series (immunity tests)

Thank you • Build something this week.